

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Title – Machine Predictive Maintenance Classification using IBM Watsonx AutoAI

Domain of Project – Artificial Intelligence | Machine Learning | Industrial Automation | Industry 4.0

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Problem Statement -

Problem Statement 39: Predictive Maintenance of Industrial Machinery

The Challenge

Unexpected machine failures in industrial environments cause:

- Production downtime
- Increased maintenance costs
- Equipment damage
- Reduced operational efficiency

Conventionally used maintenance strategies may be ineffective in predicting machine failures.

The Objective

Develop a predictive maintenance system that utilizes AI technology to:

- Analyzes machine sensor data
- Predicts machine failure categories
- Identifies risk patterns
- Supports proactive maintenance planning
- Reduces downtime and operational costs

Technology Used

Tech Stack Used

IBM Technologies

- IBM Watsonx AutoAI
- IBM Cloud Lite
- IBM Watson Machine Learning

Machine Learning Libraries

- Scikit-Learn
- Snap ML
- XGBoost
- LightGBM

Development Tools

- Python 3.12
- Jupyter Notebook
- GitHub

Dataset Source

Machine Predictive Maintenance Classification Dataset (Kaggle)

Proposed Solution -

Proposed Solution

The proposed solution is an AI-powered predictive maintenance system developed using IBM Watsonx AutoAI.

The system:

1. Collect machine sensor data
2. Preprocess and engineer features
3. Train AutoAI models
4. Predict machine failure type
5. Deploy as a real-time API

The model predicts:

- Tool Wear Failure
- Heat Dissipation Failure
- Power Failure
- Overstrain Failure
- Random Failure
- No Failure

The best model achieved:

99.5% Accuracy

using IBM AutoAI ensemble techniques.

IBM AutoAI Workflow:-

Step 1

Dataset Upload

Step 2

Data Preprocessing

Step 3

AutoAI Training

Step 4

Model Selection

Step 5

Deployment

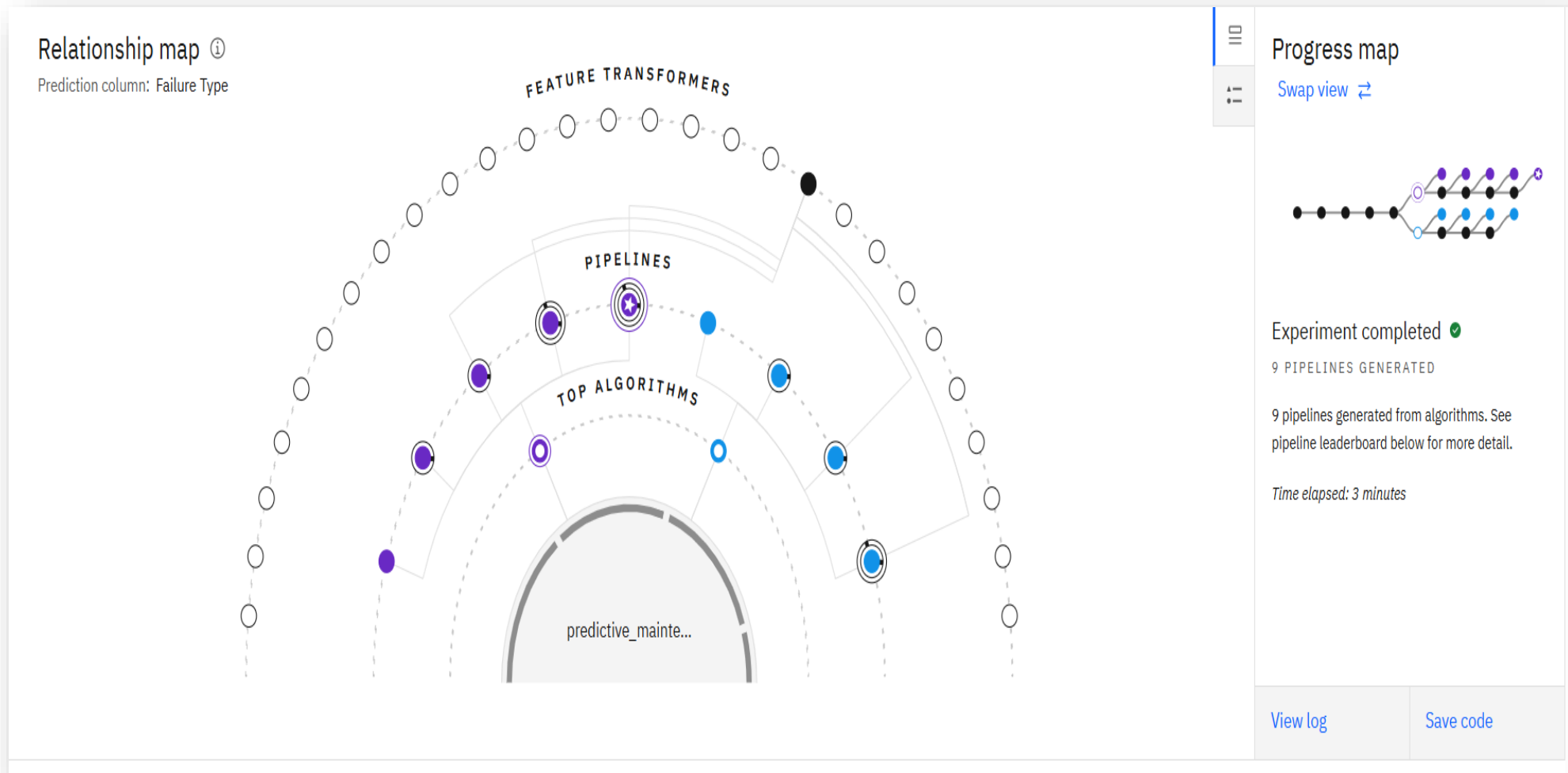
Step 6

Pipeline Evaluation

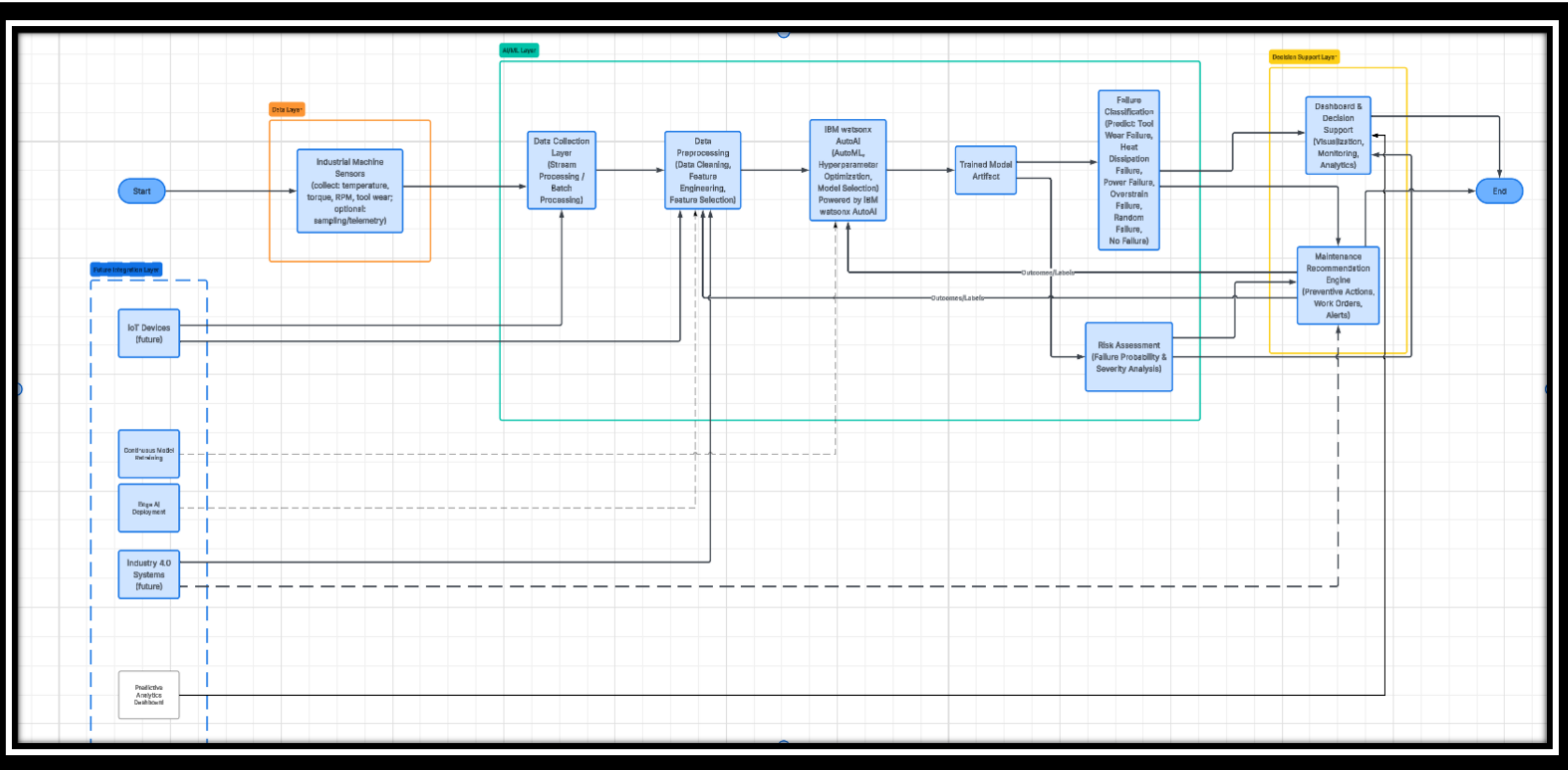
Step 7

Prediction

AutoAI workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Artificial Intelligence

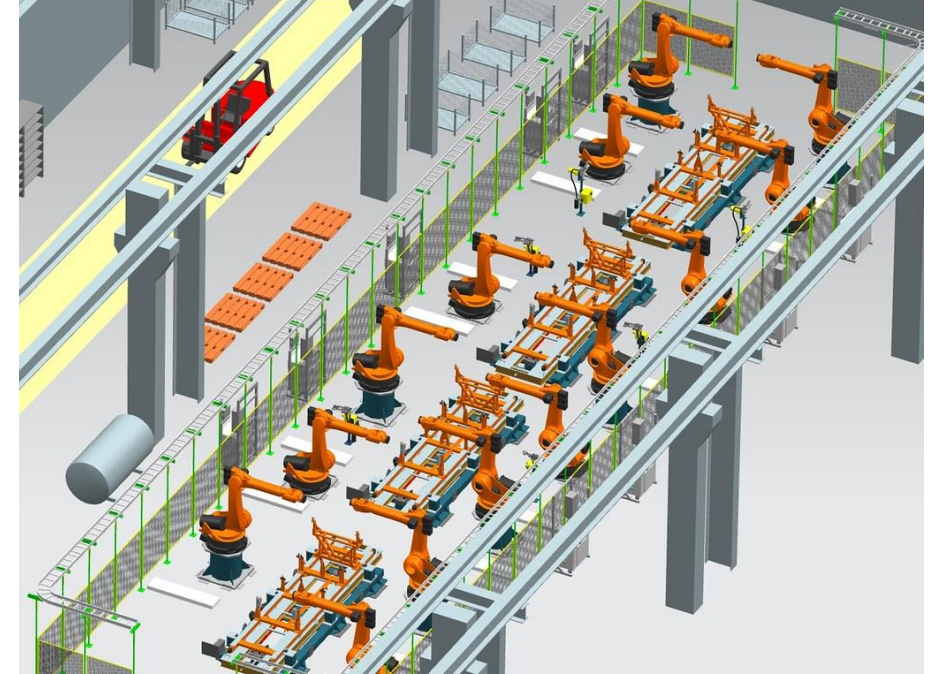
Artificial Intelligence enables:

- Automated pattern recognition
- Failure prediction
- Risk assessment
- Data-driven maintenance planning

Benefits:

- Reduced downtime
- Increased reliability
- Lower maintenance cost
- Improved productivity

The Agentic AI system acts as an intelligent maintenance assistant by monitoring machine health, predicting failures, assessing risks, and recommending corrective actions with minimal human intervention.



Technology Used

- **IBM Watsonx AutoAI**

Automated machine learning model generation and optimization.

- **IBM Watson Machine Learning**

Model management and deployment.

- **Python**

Data processing and experimentation.

- **Scikit-Learn**

Classification algorithms and preprocessing.

- **GitHub**

Source code management

Novelty and Uniqueness

➤ **Automated Machine Learning Using IBM Watsonx AutoAI**

While typical machine learning models involve manual model selection and optimization, this approach uses IBM Watsonx AutoAI to create and optimize several machine learning models automatically.

➤ **Predicting Machine Failure with High Accuracy**

A classification accuracy of 99.5% was recorded when training the model on machine failure data, which means the system can predict machine failure with high accuracy.

➤ **Predictive Maintenance for Industrial Machines**

Industrial machines are typically subjected to preventive maintenance measures, but sometimes, the measures are applied after a problem happens. This predictive maintenance method forecasts any future problems.

➤ **Scalable Industrial Approach**

The design may easily work together with manufacturing processes, industrial internet of things technology, and cloud computing, thus being applicable for implementation in smart factories.

Git Hub Link

·Please test and Paste the working Github URL – <https://github.com/AdityaDey01/Predictive-Maintenance-Classification>

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Future Scope

➤ Real-Time IoT Sensors Connection

The system can be linked with active industrial sensors that would help in monitoring the real-time data related to machine parameters including temperature, torque, vibration, and rotation rate, thereby making possible real-time failure prediction and alerting.

➤ Predictive Maintenance Suggestions Module

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Predicted failure type can help in suggesting automatic maintenance activities that would guide maintenance workers to find out the best corrective measures.

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➤ Web-Based Machine Health Monitoring System

A web-based monitoring dashboard for machine health status can be built using technologies like Streamlit or React.

Rank		Name	Algorithm	Specialization	Accuracy (Optimized) Cross validation	Enhancements	Build time
★	1	Pipeline 5	Batched Tree Ensemble Classifier (Snap Random Forest Classifier)	INCR	0.995	HPO-1 FE HPO-2 BATCH	00:00:48
	2	Pipeline 4	Snap Random Forest Classifier		0.995	HPO-1 FE HPO-2	00:00:44
	3	Pipeline 3	Snap Random Forest Classifier		0.995	HPO-1 FE	00:00:34
	4	Pipeline 2	Snap Random Forest Classifier		0.994	HPO-1	00:00:08

[illegible]

Prediction results

Prediction type

Multiclass classification

Prediction percentage

1
record

No Failure

Display format for prediction results

☒ Table view

☐ JSON view

Show input data

	Prediction	Confidence
1	No Failure	100%
2		
3		
4		
5		
6		
7		
8		
9		
10		

Download JSON file

Thank You!

Thank you for your time and interest.